

Day 1: Properties of Exponents

1. Simplify: $x^5 \cdot x^3 =$ _____
2. Simplify: $\frac{y^{10}}{y^4} =$ _____
3. Which expression is equivalent to $(2x^3)^4$?
A. $2x^{12}$ B. $8x^{12}$ C. $16x^{12}$ D. $16x^7$
4. Simplify and write with positive exponents: $\frac{x^3 \cdot x^{-7}}{x^{-2}} =$ _____

Day 2: Rational Exponents & Radicals

1. Convert to radical form: $x^{2/3} =$ _____
2. Convert to rational exponent form: $\sqrt[4]{x^3} =$ _____
3. What is the value of $8^{2/3}$?
A. 2 B. 4 C. 12 D. 16
4. Simplify: $\sqrt{50x^6} =$ _____ (Hint: $50 = 25 \cdot 2$)

Day 3: Add/Subtract Polynomials

1. Simplify: $(3x^2 + 5x - 2) + (x^2 - 3x + 7) =$ _____
2. Simplify: $(4x^3 - 2x^2 + x) - (x^3 + 5x^2 - 3x) =$ _____
3. Which is equivalent to $(2x^2 - 3x + 1) - (x^2 + 4x - 5)$?
A. $x^2 - 7x + 6$ B. $x^2 + x - 4$ C. $x^2 - 7x - 4$ D. $3x^2 + x + 6$

Day 4: Multiply Polynomials

1. Simplify: $3x^2(2x^3 - 5x + 4) =$ _____
2. Expand: $(x + 3)(x - 5) =$ _____
3. Expand: $(2x - 1)(x^2 + 3x - 2)$

Show your work:

Answer: _____

Day 5: Divide by Monomial + Review

1. Simplify: $\frac{12x^5 - 8x^3 + 4x^2}{4x^2} = \underline{\hspace{2cm}}$

2. Simplify: $\frac{15a^4b^3 - 10a^2b^5}{5a^2b^2} = \underline{\hspace{2cm}}$

3. **SAT/TSI:** If $a^2 = 5$, what is the value of a^6 ?

A. 15 B. 25 C. 125 D. 625

4. **SAT/TSI:** The expression $\frac{(3x^2y)^2}{9xy^3}$ is equivalent to:

A. x^3y^{-1} B. $\frac{x^3}{y}$ C. x^3y D. $\frac{x^2}{y}$

Answer Key (remove before copying): **D1:** 1. x^8 2. y^6 3. C 4. x^{-2} or $1/x^2$ **D2:** 1. $\sqrt[3]{x^2}$ 2. $x^{3/4}$ 3. B 4. $5x^3\sqrt{2}$ **D3:** 1. $4x^2 + 2x + 5$ 2. $3x^3 - 7x^2 + 4x$ 3. A **D4:** 1. $6x^5 - 15x^3 + 12x^2$ 2. $x^2 - 2x - 15$ 3. $2x^3 + 5x^2 - 7x + 2$ **D5:** 1. $3x^3 - 2x + 1$ 2. $3a^2b - 2b^3$ 3. C 4. B